

# A Complete Fourier Spectral Atlas for Constant- $f$ Rotating Shallow Water

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## Abstract

We give a global same-clock theorem for the linear rotating shallow-water equations on the doubly periodic  $2\pi$ -torus. Each lattice mode is a three-by-three skew-Hermitian block. Its exact zero, plus, and minus projectors resolve the stationary geostrophic/potential-vorticity branch and the two inertia-gravity branches, including all zero-parameter faces. We also prove the sum-of-two-squares shell multiplicities, a necessary and sufficient periodicity test for finite Fourier support, and the global unitary but noncompact/non-Schatten boundary. A deterministic ledger and independent symbolic and numerical paths audit the claims. The model has constant  $f$ ; no beta-plane Rossby assertion is made.

## 1 Model and statement

Let  $u = (u_1, u_2)$  and  $\phi$  be complex fields on  $\mathbb{T}^2$ , and let

$$u_t + fJu + c\nabla\phi = 0, \quad \phi_t + c\nabla \cdot u = 0, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here  $f \in \mathbb{R}$ ,  $c \geq 0$ , and  $t$  is physical time. With  $e_n(x) = e^{in \cdot x}$ ,  $n \in \mathbb{Z}^2$ , the coefficient vector  $X_n = (\hat{u}_{1,n}, \hat{u}_{2,n}, \hat{\phi}_n)^T$  obeys

$$\dot{X}_n = G_n X_n, \quad G_n = \begin{pmatrix} 0 & f & -icn_1 \\ -f & 0 & -icn_2 \\ -icn_1 & -icn_2 & 0 \end{pmatrix}, \quad \omega_n = (f^2 + c^2|n|^2)^{1/2}.$$

For  $c > 0$ , the Fourier multiplier is taken on its maximal graph domain

$$D(G) = \left\{ X \in L^2 : \sum_{n \in \mathbb{Z}^2} \|G_n X_n\|_{\mathbb{C}^3}^2 < \infty \right\};$$

trigonometric polynomials form a core and  $H^1$  is contained in this domain. The skew-Hermitian blocks close to a skew-adjoint generator and hence to the global unitary  $L^2$  group. When  $c = 0$ , the bounded pointwise rotation is treated separately.

**Theorem 1** (projector, shell, and operator atlas). *For every  $n$  with  $\omega_n > 0$ ,  $G_n^* = -G_n$ ,*

$$G_n^3 + \omega_n^2 G_n = 0, \quad P_0 = I + \frac{G_n^2}{\omega_n^2}, \quad P_{\pm} = \frac{1}{2} \left( -\frac{G_n^2}{\omega_n^2} \mp \frac{iG_n}{\omega_n} \right),$$

*and  $P_0, P_+, P_-$  are mutually orthogonal projectors with ranks 1, 1, 1. Moreover*

$$e^{tG_n} = P_0 + \cos(\omega_n t)(I - P_0) + \frac{\sin(\omega_n t)}{\omega_n} G_n.$$

The  $P_0$  branch is stationary and is the linear potential-vorticity/geostrophic branch;  $P_{\pm}$  are the inertia-gravity branches. At  $n = 0$ ,  $f \neq 0$  gives  $0, \pm i|f|$ , while  $f = 0$  gives a zero block. For  $c > 0$ , the Fourier potential-vorticity quantity

$$\hat{\zeta}_n - \frac{f}{c}\hat{\phi}_n, \quad \hat{\zeta}_n = i(n_1\hat{u}_{2,n} - n_2\hat{u}_{1,n}),$$

is constant in time.

For  $q > 0$ , the shell  $|n|^2 = q$  has  $r_2(q) = 4(d_1(q) - d_3(q))$  modes. A finite Fourier-support solution is  $T$ -periodic if and only if  $\omega_n T \in 2\pi\mathbb{Z}$  for every participating nonzero branch. The full group on  $L^2(\mathbb{T}^2; \mathbb{C}^3)$  is unitary, but for every  $t$ , including  $t = 0$ , it is neither compact nor in any Schatten class.

*Proof.* Fourier transformation gives  $G_n$ . Direct multiplication yields the minimal polynomial  $z(z^2 + \omega_n^2)$ , and inspection of the entries gives  $G_n^* = -G_n$ . The three Lagrange polynomials for the roots  $0, \pm i\omega_n$  are precisely  $P_0, P_+, P_-$ , proving orthogonality and the exponential after reducing its power series modulo the cubic. Taking  $\zeta = \partial_1 u_2 - \partial_2 u_1$  gives  $\zeta_t = -f\nabla \cdot u$ , while  $\phi_t = -c\nabla \cdot u$ , proving the potential-vorticity identity for  $c > 0$ . The shell formula is the classical sum-of-two-squares identity. Applying the projectors to a finite superposition proves the periodicity equivalence. Finally, skew-adjointness gives unitarity; an orthonormal sequence of distinct Fourier modes is carried to another orthonormal sequence of norm one, proving noncompactness and the failure of every Schatten condition.  $\square$

## 2 Degeneracies and finite controls

The faces  $f = 0$ ,  $c = 0$ , and  $n = 0$  are not silently folded into the generic projector formula. In particular,  $f = c = 0$  is a three-dimensional zero block, whereas  $c = 0$  leaves a scalar stationary component at every mode. The ledger contains eight rate cases, including a negative- $f$  retrograde sentinel, and all integer modes in  $[-3, 3]^2$ . It also checks the shell identity for  $q \leq 18$ . The periodicity assertion is only for finite support; an arbitrary  $L^2$  state need not be periodic.

## Source note

The  $f$ -plane doubly periodic normal-mode context is discussed by Mohebalhojeh and Dritschel [1]; Salmon [2] provides Hamiltonian projection context. Beta-plane results are intentionally outside the frozen model.

## References

- [1] A. R. Mohebalhojeh and D. G. Dritschel, *Hierarchies of Balance Conditions for the  $f$ -Plane Shallow-Water Equations*, *Journal of the Atmospheric Sciences* 58 (2001), 2411–2426. DOI: 10.1175/1520-0469(2001)058<2411:HOBCT>2.0.CO;2.
- [2] R. Salmon, *Semigeostrophic theory as a Dirac-bracket projection*, *Journal of Fluid Mechanics* 196 (1988), 345–358. DOI: 10.1017/S0022112088002733.

## Declarations

**Scope.** NO\_BAD\_EULER\_OR\_ROOT\_NUMBER; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C217. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.